Multivariate Calculus Lecture Notes
Section 16.5 – Surface Integrals of Vector Fields (short version)

I. Idea
   a. Suppose we want to calculate the flux of a vector field across some 3D surface S, like so:
      i. This can be interpreted as the flow rate of some fluid across the surface S.
      ii. Then we need some sort of vector surface integral.
   b. To calculate this, we would need to find how much of the vector field $\mathbf{F}$ aligns with some outward-pointing unit normal vector $\mathbf{n}$, as $\mathbf{n}$ roams across the surface S.
   c. So we define the integral of the vector field $\mathbf{F}$ over the surface S, or *vector surface integral*, to be
$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S (\mathbf{F} \cdot \mathbf{n})\, dS.$$

II. Calculating a vector surface integral
   a. As with a line integral, to calculate a vector surface integral we need a parameterization $G(u, v)$ of the surface.
      i. A common paramaterization of the sphere of radius ρ is given by spherical coordinates $G(\theta, \phi) = (\rho \cos\theta \sin\phi, \rho \sin\theta \sin\phi, \rho \cos\phi)$.
      ii. See Section 16.4 for details.
   b. To get a normal vector $\mathbf{N}(u, v)$, we can take the cross product of the tangent vectors $\mathbf{T}_u(u, v)$ and $\mathbf{T}_v(u, v)$ to the surface:
   c. Thus $\mathbf{N} = \mathbf{T}_u \times \mathbf{T}_v$ or $\mathbf{N} = -\mathbf{T}_u \times \mathbf{T}_v$, depending on the orientation.
   d. It turns out (see Section 16.4) that $dS = \|\mathbf{N}(u, v)\|\, du\, dv$.
   e. This gives us
$$\begin{aligned}\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S (\mathbf{F} \cdot \mathbf{n})\, dS &= \iint_D \mathbf{F}(G(u, v)) \cdot \frac{\mathbf{N}(u, v)}{\|\mathbf{N}(u, v)\|} \|\mathbf{N}(u, v)\|\, du\, dv \\ &= \iint_D \mathbf{F}(G(u, v)) \cdot \mathbf{N}(u, v)\, du\, dv,\end{aligned}$$
      where D is the domain of the surface in terms of the u-v parameter space.
   f. Note that the Jacobian has already been included in the integral expression; we won't need to determine it or add it in explicitly when evaluating one of these integrals.

III. Example: Find the outward flux of the vector field $\mathbf{F}(x, y, z) = y\mathbf{i} + x\mathbf{j} + z\mathbf{k}$ across the surface S, where S is the graph of $z = 1 - x^2 - y^2$ for $z \geq 0$.
   a. First, we parameterize the surface.
      i. The restriction $z \geq 0$ means that $1 - x^2 - y^2 \geq 0$, or $x^2 + y^2 \leq 1$.

ii. The easiest way to parameterize this is in polar coordinates: $x = r\cos\theta$, $y = r\sin\theta$, $0 \le r \le 1$, $0 \le \theta \le 2\pi$.

iii. So we have the parameterization $G(r,\theta) = (r\cos\theta, r\sin\theta, 1-r^2)$, for $0 \le r \le 1$, $0 \le \theta \le 2\pi$.

b. Then, find $\mathbf{N}$.

i. $\mathbf{T}_r = \cos\theta\mathbf{i} + \sin\theta\mathbf{j} - 2r\mathbf{k}$

ii. $\mathbf{T}_\theta = -r\sin\theta\mathbf{i} + r\cos\theta\mathbf{j}$

iii. Thus

$$\mathbf{N} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \cos\theta & \sin\theta & -2r \\ -r\sin\theta & r\cos\theta & 0 \end{vmatrix} = (2r^2\cos\theta)\mathbf{i} + (2r^2\sin\theta)\mathbf{j} + r\mathbf{k}.$$

iv. We know this is outward-pointing because the positive $\mathbf{k}$ component means $\mathbf{N}$ always points up.

c. Next, $\mathbf{F}(G(r,\theta)) = r\sin\theta\mathbf{i} + r\cos\theta\mathbf{j} + (1-r^2)\mathbf{k}$.

d. Then $\mathbf{F}\cdot\mathbf{N} = 2r^3\sin\theta\cos\theta + 2r^3\sin\theta\cos\theta + r - r^3 = 4r^3\sin\theta\cos\theta + r - r^3$.

e. Therefore,

$$\iint_S \mathbf{F}\cdot d\mathbf{S} = \int_0^{2\pi}\int_0^1 (4r^3\sin\theta\cos\theta + r - r^3)\,dr\,d\theta = \int_0^{2\pi}\left[r^4\sin\theta\cos\theta + \frac{r}{2} - \frac{r^4}{4}\right]_0^1 d\theta$$

$$= \int_0^{2\pi}\left(\sin\theta\cos\theta + \frac{1}{4}\right)d\theta = \frac{\sin^2\theta}{2} + \frac{\theta}{4}\Big|_0^{2\pi} = \frac{\pi}{2}.$$

IV. Applications (see text)

a. Fluid flux through a surface

b. Faraday's Law of Induction, which features both an electric field and a related magnetic field

Homework: Section 16.5 #2, 5, 6, 7, 9, 10